

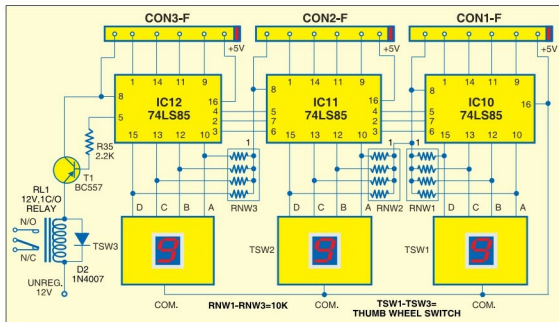


Fig. 3: Comparator unit of digital temperature controller

is connected to input pin 10 of ADC ICL7135 (IC2).

Analogue-to-digital converter (ADC). ICL7135 is a 4½-digit ADC. This dual-slope integrating ADC provides interface for the visual display. ICL7135 is available in a 24-pin plastic dual-in-line package (PDIP). The output of signal amplifier op-amp $\mu A741$ is fed to input pin 10 of ADC ICL7135 to get multiplexed binary-coded decimal (BCD) outputs B1 through B4 and digits D1 through D3 from pins 13 through 16 and pins 20 down through 18, respectively. Pin 22 of ICL7135 is connected to the output of IC NE555 (IC3).

IC NE555 is configured as a free running oscillator for multiplexing of the digital output inside ADC ICL7135. Preset VR4 is used to set the frequency of oscillations at 120 kHz. Latches 74LS75 (IC4 through IC6) are used to get stable, individual unit's, ten's and hundred's digits from three-digit multiplexed output of IC 7135 for displaying the actual temperature through three 7-segment display units. At the same

time, the temperature comparator unit compares the actual temperature value with the preset value, to energise relay RL1 to control the heating element of the oven or furnace. A connector is provided at the output of latches 74LS75 (IC4 through IC6) to interface other digital instruments for further temperature study, temperature recording, temperature processing, data logging and data storage on the hard disk of a PC.

Temperature display unit. The three-digit BCD output of latches (IC4 through IC6), representing unit's, ten's and hundred's digits, is fed to the tem-

perature display unit as an input for BCD-to-7-segment decoder 74LS48 (IC7 through IC9). Three 7-segment displays are used to display the actual temperature in three decimal digits (up to 999°C).

Digital comparator unit.

Fig. 3 shows the circuit of the digital comparator unit. The three-digit BCD output of 74LS75 (IC4 through IC6) is also fed to the digital comparator unit. Three 4-bit 74LS85 magnitude comparators are used here for comparison of the actual temperature with the preset temperature value. Three BCD thumbwheel switches

TWS1 through TWS3 are used for temperature setting. If the actual temperature value is higher than the preset temperature value, pin 5 of the most significant digit comparator (IC12) goes low. This drives transistor T1 into saturation and relay RL1 energises to control the heating element of the oven.

Power supply unit. Fig. 4 shows the circuit of the power supply. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 9V-0-9V, 1A. The transformer output is rectified by a full-wave rectifier comprising diodes D3 through D6,

filtered by capacitors C11 and C12, and regulated by ICs 7805 (IC13) and 7905 (IC14) to get regulated +5V and -5V, respectively. Capacitor combinations C13-C15 and C14-C16 bypass ripples, if any, in the positive and negative power supplies, respectively. LED1 acts as the power indicator and R36 limits the current through LED1.

Construction

An actual-size, single-side PCB for the digital tem-

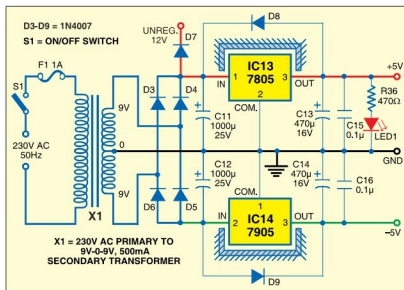


Fig. 4: Power supply